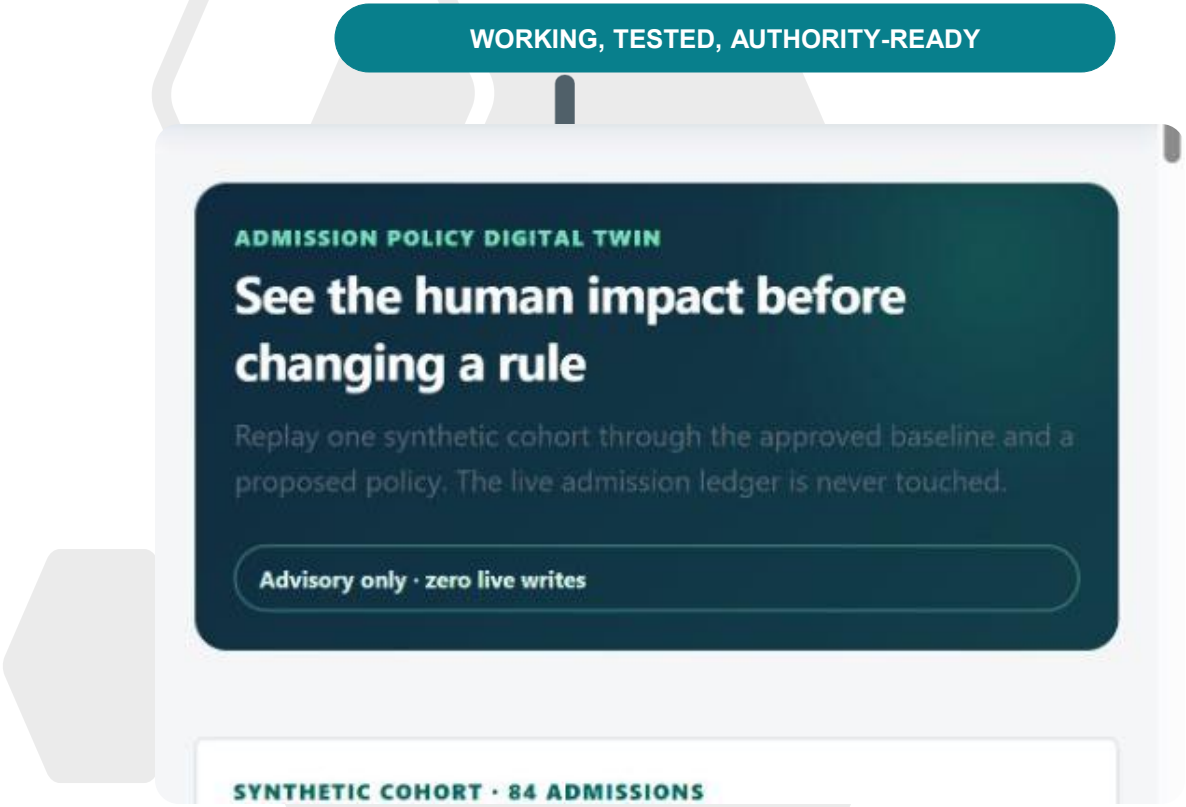
SMART INDIA HACKATHON 2026

ADMISSION REGISTRY

Problem Statement ID 26263

- APAAR ID-Based Student Data Management & Verification
- Theme Smart Automation
- Category Software
- Team ID [TEAM ID]



Protect every legitimate seat before admission becomes an audit problem.

[TEAM NAME]

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IDEA TITLE

FROM ANNUAL TOTALS TO A VERIFIABLE ADMISSION LEDGER

TODAY'S FAILURE CHAIN

Fragmented files

Student identity gaps

Late intake reconciliation

Duplicate or excess enrolment discovered after the fact

1. **Versioned upload**
2. **APAAR adapter**
3. **Policy decision**
4. **Seat protected**
5. **Immutable receipt**
6. **AICTE oversight**

THE DIFFERENCE

Last-seat concurrency protection • Every outcome has a rule code • Policy changes can be replayed before rollout

HERO FEATURE: ADMISSION POLICY DIGITAL TWIN

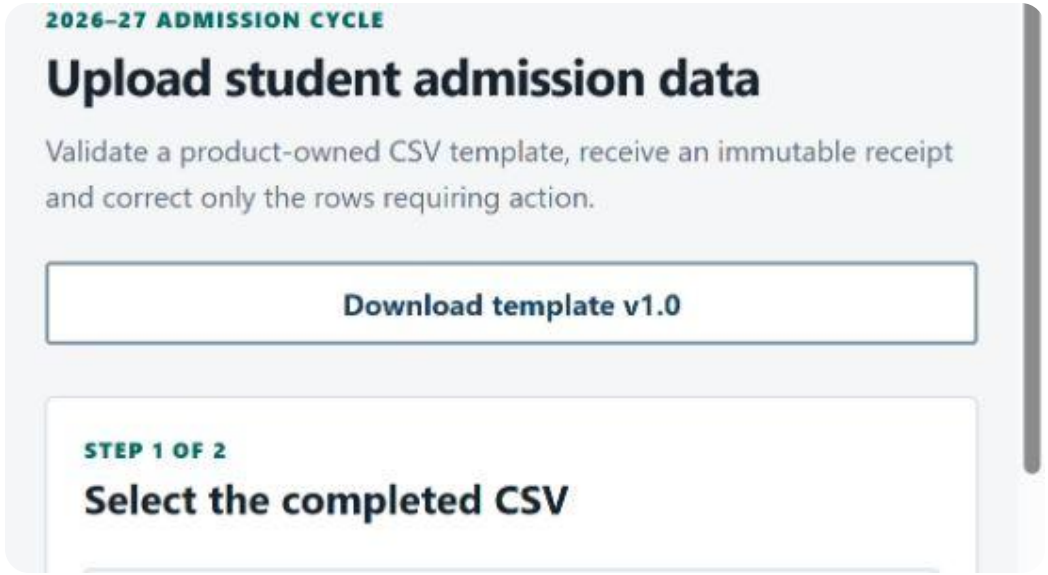
[TEAM NAME]

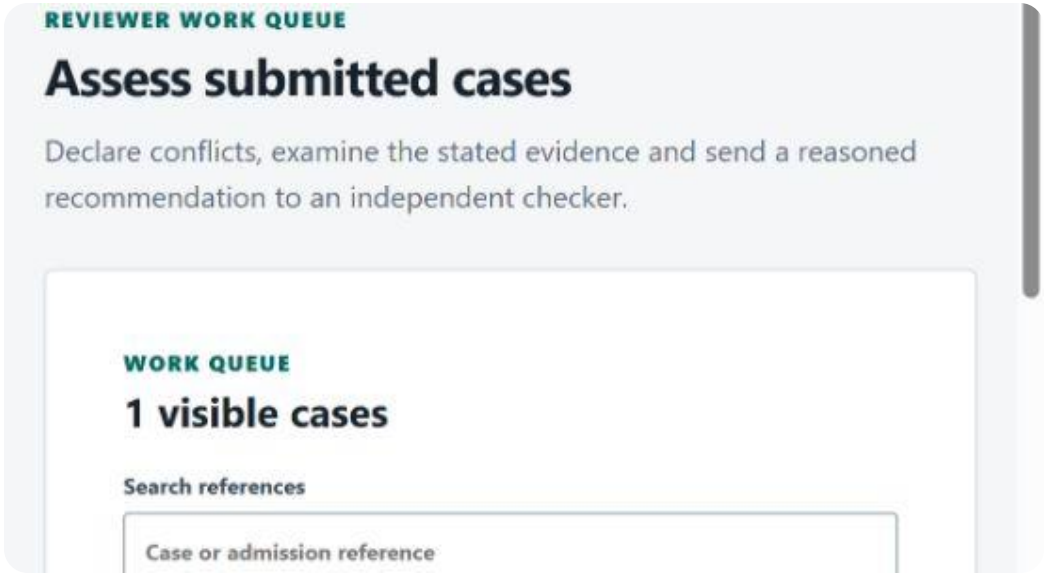
TECHNICAL APPROACH

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ONE POLICY ENGINE. ONE TRANSACTION. ONE DEFENSIBLE RECORD.

7 ROLES — Institution + AICTE → **GATEWAY** — OIDC/RBAC boundary → **DOMAIN** — Admission + batch + casework → **POSTGRESQL** — Serializable seat lock → **ADAPTERS** — APAAR/SSO/signing





REAL POSTGRESQL CONCURRENCY | **RESTART PERSISTENCE** | **7-ROLE ACCESS TESTING**

[TEAM NAME]

FEASIBILITY AND VIABILITY

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BUILT & TESTABLE NOW

Policy kernel + category capacity
Versioned CSV ingestion + correction
Transactional audit/outbox + idempotency
Governed casework + seven scoped roles
Containers, recovery, security and accessibility checks

AUTHORITY-CONTROLLED NEXT

Official APAAR/API Setu credentials
AICTE SSO, approval feed and signing service
Production hosting, keys, SLO/RTO/RPO
Privacy, legal and security approvals
Representative supervised pilot cohort

RISK → CONTROL

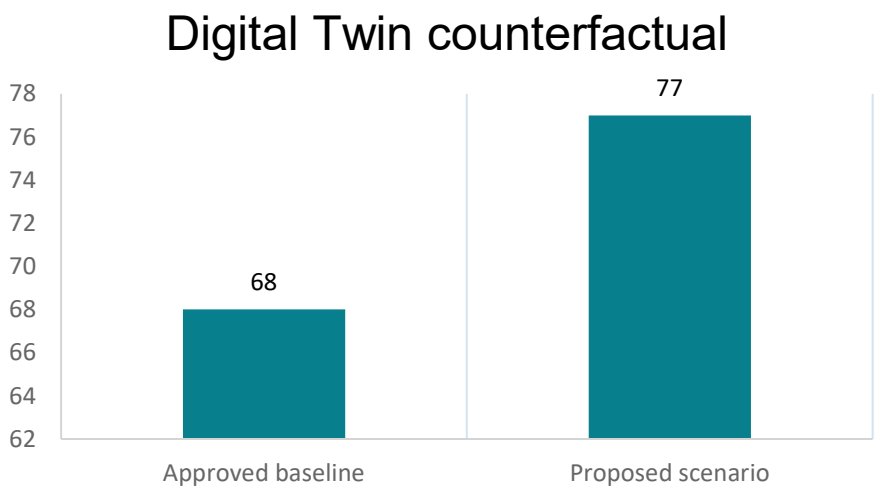
Policy ambiguity	→	**Manual review; never invent authority**
Provider outage	→	**Explicit outage state + safe retry**
Duplicate / final seat	→	**Idempotency + serializable lock**
Privacy leakage	→	**Data minimisation + scoped audit**

[TEAM NAME]

IMPACT AND BENEFITS

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PREVENT HARM TODAY. IMPROVE POLICY BEFORE TOMORROW.



+9 policy-valid access

5 excess still blocked

Scenario evidence only — synthetic cohort, deterministic rules, zero live writes.



STUDENT	**Explainable identity-linked record**	**INSTITUTION**	**Row correction + defensible receipt**
AICTE	**Intake integrity + policy foresight**	**VERIFIER**	**Purpose-limited audit trail**

[TEAM NAME]

RESEARCH AND REFERENCES

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AUTHORITATIVE RULES + EXECUTABLE PROOF

PRIMARY SOURCES

1 AICTE Approval Process Handbook 2024–2027
aicte-india.org

2 APAAR About, Privacy Policy and Terms
apaar.education.gov.in

3 API Setu ABC / APAAR catalogue + consumer SOP
directory.apisetu.gov.in · apisetu.gov.in

4 SIH 2026 format and evaluation guidance
sih.gov.in

REPOSITORY EVIDENCE

✓ Domain rule corpus + stable rule codes
✓ Real PostgreSQL migration/concurrency suites
✓ Batch, casework and integration contracts
✓ Full-system restart-persistence journey
✓ Seven-role accessibility checks
✓ Container, recovery, audit and secret scans
✓ SBOM and recursive licence inventory
✓ Synthetic identifiers only

JUDGE GUIDE + REPOSITORY QR WILL BE ADDED AFTER GITHUB URL IS SET